

Building Legal RAG: what each component contributes

01 CONTEXT AND HYPOTHESIS

Retrieval-augmented generation (RAG) grounds language-model answers in external documents [1]. Legal QA needs correct answers and precise, complete source-page citations [2, 3]. PDFs may lack searchable text; evidence spans different text scales and may require exact legal terms. Retrieving more passages can also introduce irrelevant pages.

How can a legal RAG pipeline improve answer quality while keeping source-page citations both precise and complete?

H1 · Retrieval Multiple text scales, hybrid search and reranking will improve answers and evidence retrieval over R0.	H2 · Scan robustness Recovering text from scanned documents will reduce the performance loss caused by the missing text layer.	H3 · Citation selection Selecting pages after answer generation will improve citation precision while retaining most of the relevant evidence.
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02 METHODOLOGY

Data: 30 DIFC legal PDFs (590 pages), 100 questions: 70 structured-answer and 30 free-text. Reference source pages are available for 95 questions. In the mixed corpus, 10 PDFs (107 pages) become synthetic scans: page images at 300 DPI with the text layer removed.

R0 · Baseline. R1–R6 each include all earlier additions. PDF text → 300-token passages → semantic retrieval → GPT-5.4-mini → answer + retrieved pages as citations. Semantic search uses Qwen3-Embedding-0.6B.

Document processing		Retrieval		Answering	Citations
R1 · Read scans Recover searchable text from page images. OCR: PP-OCRv5	R2 · Multiple text scales Represent evidence at different levels. Pages, sections, clauses, microchunks, tables	R3 · Find exact terms Combine semantic and lexical search. BM25 + rank fusion (RRF)	R4 · Rank evidence Prioritize the most relevant passages. Qwen3-Reranker-0.6B	R5 · Check answer type Normalize and validate the expected type. Type checks, confidence thresholds	R6 · Refine citations Select pages linked to supporting passages. Generator's supporting-chunk references

Fixed: GPT-5.4-mini / high, 24 retrieval candidates, ≤ 2,400 context tokens and the generation schema. R4–R6 reuse one generated response. R0–R6 run on the mixed corpus; R0/R1 also on originals.

Evaluation: Correct answers can still have poor citations. We score answers and citations separately.

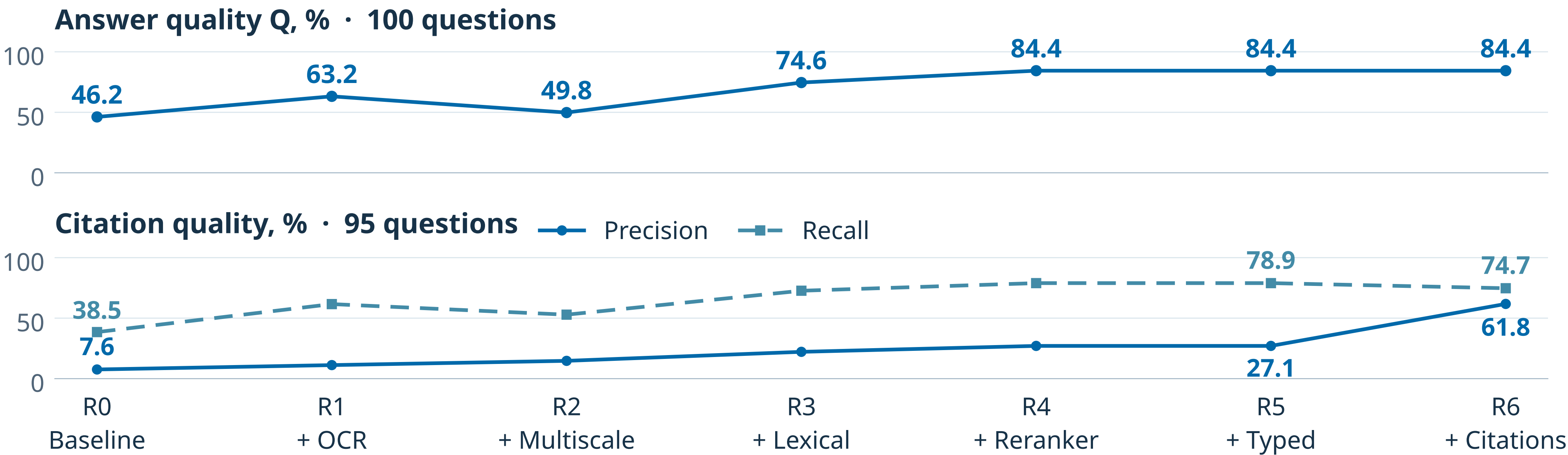
Answer quality $Q = 0.7 \times S + 0.3 \times F$. S / F: mean structured / free-text scores. Free text is LLM-scored.

Citation precision: share of cited pages that match annotated source pages. Are the citations relevant?

Citation recall: share of annotated source pages cited. Is the evidence complete? Both average 95 questions.

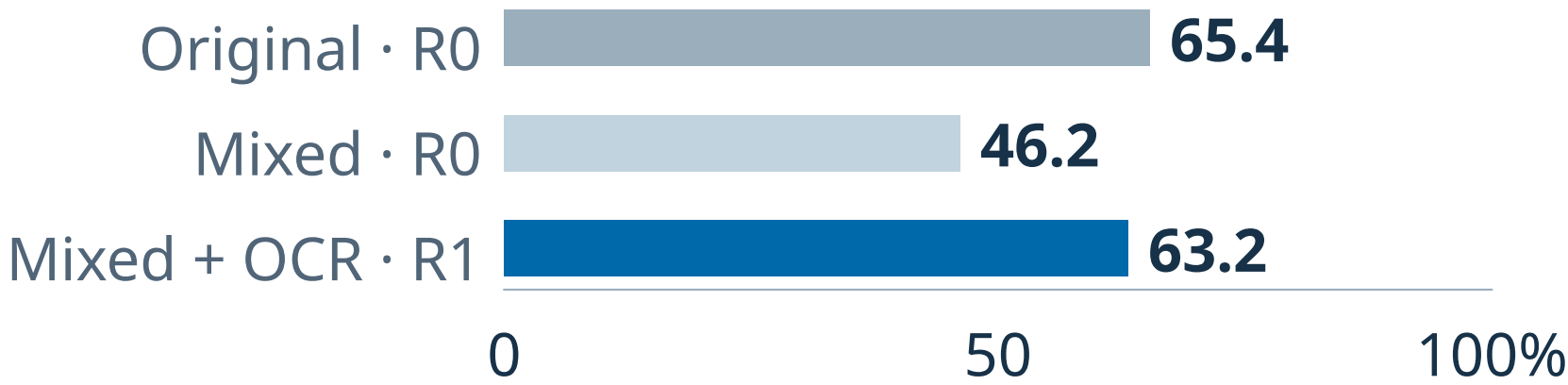
03 RESULTS

R0 to R6 on the mixed corpus



Q is an aggregate score, not the share of fully correct answers. R5 adds no score gain.

OCR recovers performance on scans



Scanning 10 of 30 PDFs lowers Q from 65.4% to 46.2%. OCR restores most of this loss, reaching 63.2%.

Hypothesis evaluation

- H1 · Partially supported.** Multiple scales alone reduce Q; lexical retrieval and reranking improve Q and recall.
- H2 · Supported.** OCR restores most of the loss caused by removing the text layer.
- H3 · Supported with a trade-off.** Precision rises from 27.1% to 61.8%; recall falls from 78.9% to 74.7% (−4.2 pp).

Conclusion
OCR recovers text from scans, hybrid retrieval and reranking improve answers, and citation selection trades recall for precision. Evaluate and optimize answer quality and citations separately.

Limitations: one DIFC development corpus, 100 questions and synthetic scans; sequential additions; one run per condition, no significance testing; LLM-scored free text.

References: [1] Lewis et al. RAG (2020). [2] Pipitone & Alami. LegalBench-RAG (2024). [3] Gao et al. ALCE (2023).

Code, protocol & results: github.com/SteinVR/Trends-in-NLP-Poster